2

Sex Hormones and Modulation of Immune Response in SLE

NORMAN TALAL

SLE and most other autoimmune diseases occur more commonly in females. The female: male ratio in SLE can be as high as 10:1 when considering women in their menstruating years. This ratio falls to 3:1 when considering non-menstruating girls or post-menopausal females (Kornreich, 1976). Furthermore, clinical improvement in SLE has been reported following removal of the female sex organs (Yocum et al, 1975; Rose and Pillsbury, 1944). Conversely, SLE may exacerbate during pregnancy (Chapel and Burns, 1971) or in association with oral contraceptives (Mund, Simson and Rothfield, 1963). SLE also develops in patients with Klinefelter's syndrome, an XXY condition in which males develop gynecomastia, sparse beard and other feminizing features. These are important clues to a possible role of sex factors in the pathogenesis of SLE.

Our laboratory became interested in the role of sex factors in autoimmune disease through studies seeking an explanation for this marked female predominance of SLE. At the time, we were studying the laboratory model of SLE that develops spontaneously in NZB/NZW (B/W) F_1 mice. These mice are genetically susceptible to a lupus-like syndrome characterized by the formation of antibodies to nucleic acids, immune complex glomerulonephritis and death from uraemia. Female mice develop this illness several months earlier than males. We thus undertook a series of investigations in B/W mice to study the effects of sex steroid hormones on autoimmune disease.

SEX HORMONE MODULATION OF MURINE LUPUS

In our first experiment, we removed the gonads from B/W mice prior to puberty and observed that castrated males developed an accelerated disease pattern identical to that of females (Roubinian, Papoian and Talal, 1977). They died at the same rate as females. Moreover, castration of females had no effect on disease expression, that is, there was no obvious benefit from oophorectomy.

Clinics in Rheumatic Diseases—Vol. 8, No. 1, April 1982
0307-742X/82/0801-23 $03.00

These results suggested a protective influence of the male gonad, presumably due to production of androgenic hormone. Accordingly, we administered male hormone to female B/W mice and observed marked amelioration of their disease (Roubinian et al, 1978). Continuous administration of androgen allowed these female lupus-susceptible mice to live a normal life span. This therapeutic effect was observed even when male hormone was delayed to an age when disease was already established (Roubinian et al, 1979b).

The aetiology of murine lupus is multifactorial and includes genetic, hormonal, immunological and possibly viral components. Likewise, the immunoregulatory abnormalities in the autoimmune mice are multiple and involve T cells, B cells, macrophages and probably other cells as well. Some background information on these abnormalities will be helpful before considering where sex steroid hormones may act to influence disease expression.

There has been a great deal of interest and some degree of controversy about which cell is involved primarily in B/W mice. The initial culprit was thought to be the T cell. This idea was supported by the discovery of natural thymocytotoxic autoantibody in lupus serum which has the ability to kill suppressor T cells. This autoantibody contributes to the deficiency of suppressor T cells which exists in human and murine lupus. How important this mechanism might be in vivo is still uncertain. Additional evidence for T cell abnormalities relates to the deficiency of thymic hormones which develops in NZB and B/W mice.

In recent years, much evidence has also accumulated to suggest that B cell abnormalities arise primarily and not simply as a consequence of suppressor T cell deficiency. Lupus mice behave as if under the influence of intrinsic polyclonal B cell activators. This seems analogous to the influence of lipopolysaccharide in normal mice.

This controversy between the T cell and B cell schools is probably resolvable when one considers that the immune system is a network of interacting components. In all likelihood, the primary defect is in a stem cell; this defect is then transmitted into both T and B cell lineages. Furthermore, the defect in the macrophages could be even more important than the T and B cell abnormalities. Regulation of the immune response by macrophages expressing Ia on the cell surface is an early and critical event in the network of immunological control.

We have performed many experiments attempting to determine which cells are affected by sex steroid hormones. Two possible sites of action are macrophages and the thymus.

With Dr Hannah Shear from New York University (Shear et al, 1982) we have studied the clearance in mice of IgG-coated autologous erythrocytes. Lupus patients show delayed clearance of such particulate immune complexes. Likewise, B/W mice have a delayed clearance which becomes abnormal first in females. Clearance in female B/W mice can be improved by the administration of androgen; in males, it is made worse by the administration of oestrogen. Presumably these effects on clearance are mediated by actions of sex steroid hormones on macrophages and other cells of the reticuloendothelial system. Thus, the macrophage may be an important target site for sex hormones.

We have also obtained evidence for sex steroid hormone receptors in mouse thymus (Talal et al, 1981). We observed oestrogen-receptor-like binding in the cytoplasm prepared from thymus glands obtained from both male and female mice of several strains including Balb/c, C57BL, NZB and B/W. The observed binding was of high affinity ($K_D = 10^{-10} - 10^{-9}$mol/l) and low capacity and was specific, as is characteristic of receptor binding. Furthermore, oestradiol binding was inhibited by diethylstilboestrol but not by dihydrotestosterone, progesterone or cortisol. Treatment of castrated B/W mice with oestradiol resulted in translocation of the cytoplasmic receptor to the nucleus. We also found specific binding of dihydrotestosterone in thymic cytoplasm from Balb/c, NZB and B/W mice. Thus, there appear to be receptors for sex steroid hormones in the thymus from several strains of mice, both normal and autoimmune. The binding affinity and steroid specificity of these receptors are consistent with tissue oestrogen receptors found in the uterus of rats and other species.

We do not know whether these receptors are on thymocytes or on thymic epithelium. There is autoradiographic evidence for nuclear concentration of [^{3}H]oestradiol in thymic reticular cells of rats (Stumpf and Sar, 1976). We suspect therefore, that they are on thymic epithelium, the endocrine component of the thymus. Such a model would create an endocrine circuit in which gonadal products might influence thymic hormone production. Furthermore, if sex steroids act on macrophages and thymic epithelium, then their ability to influence lymphocyte activity could be indirect rather than direct.

A severe form of lupus also develops spontaneously in MRL/1pr mice; this is related to the presence of the *1pr* gene (Murphy and Roths, 1978). This lupus model is associated with massive proliferation of a T cell subset which has the characteristics of helper T cells (Theofilopoulos et al, 1979; Sawada and Talal, 1979; Theofilopoulos et al, 1980). Androgen treatment of MRL/1pr mice results in prolonged survival and reduced anti-DNA antibody concentration (Steinberg et al, 1980).

The effects of sex hormones on autoimmunity can also be demonstrated in a rat thyroiditis model. Testosterone treatment reduces the incidence of lesions and the levels of anti-thyroglobulin antibodies (Ansar and Penhale, 1980). Thus, the ability of androgen to suppress two different forms of autoimmune disease (one generalized, the other target-organ-specific) suggests the probable uncovering of some general aspect of hormone–immune interaction.

STUDIES IN HUMAN SLE

Experimental work to identify a role for sex factors in human antoimmunity really began with the observation in patients with SLE and Klinefelter's syndrome of chronic oestrogenic stimulation due to elevated oestradiol levels by Stern et al (1977). These authors then showed that both male and female patients with SLE had elevated conversion of oestradiol to 16-hydroxylated metabolites (Lahita et al, 1979). The significance of increased 16-hydroxylation of oestradiol is related to the ability of these oestrogenic metabolites to bind to oestrogen receptors and thus behave as active oestrogens. These results suggest that SLE patients may be under a hyperoestrogenic influence due to an alteration in oestrogen metabolism.

DISCUSSION

The interactions between the immune and endocrine systems revealed by studies in autoimmune patients and mice should be seen in the broad context of a field that might be called immunoendocrinology (Talal, 1979), representing an interface between the otherwise separate disciplines of immunology and endocrinology. Indeed, the effects of sex steroid hormones on murine lupus may represent physiological effects of hormones acting on a disordered system of immune regulation. There is considerable evidence that normal immune responses, both humoral and cellular, are greater in females than in males. This holds true both for humans and for experimental animals (Cohn, 1979).

Interactions between the immune and endocrine systems are not limited to endocrine influences on immunity but include immunological modulation of hormone receptors (for example insulin receptors), lymphocytic regulation of osteoclast activity and bone metabolism, and thymic activity as an endocrine organ. The presence of steroid-sensitive receptors in the hypothalamus raises the possibility that neuroendocrine pathways also interact with the immune system. Neuroendocrine influences superimposed upon the immune and endocrine systems would create pathways by which the brain could control certain immunological functions. Evidence is accumulating to support a role for neuroendocrine factors in immunity (Besedovsky and Sorkin, 1977; Wybran et al, 1979) which might account for reports relating stress to the development of autoimmune diseases (Rogers, Dubey and Reich, 1979).

It is interesting that thymic epithelium and reticuloendothelium may be important sites of action for sex steroid hormones. These two target tissues, although not themselves lymphoid, are enormously influential in regulating immune function. Thymic epithelium expresses Ia antigens which play a major role in directing lymphocyte specificity and in inducing helper-T cell activity. Thymic epithelium also produces hormones which are intimately involved in peripheral T cell differentiation. Thus, by influencing the expression of Ia or the production of thymic hormones, sex steroids could exert influence on the immune response at several levels. Likewise, immune complexes combine with lymphocyte or macrophage Fc receptors, fix complement and induce inflammatory events. Effects of sex steroid hormones on immune complex clearance by cells of the reticuloendothelial system would also have important immunoregulatory actions.

Our studies on sex steroid hormones represent efforts to find better and more specific ways to restore immunological regulation and turn off autoimmune processes. Already, some preliminary studies are under way attempting to evaluate endocrine manipulation as possible therapy for human autoimmune diseases. One approach involves the use of anti-oestrogens which are effective in B/W mice (Steinberg et al, 1979). However, the results to date of a double-blind study of tamoxifen do not look promising (Mackay, personal communication). Another approach uses danazol, an attenuated synthetic androgen that is effective therapy in hereditary angioneurotic oedema. Although this agent is not effective in B/W mice (Roubinian et al, 1979a), there is some evidence that lupus patients may benefit from treatment with danazol (Agnello, 1981).

Autoimmune diseases are multifactorial in nature. It is likely that different factors will play either major or minor pathogenetic roles in any individual patient (Talal et al, 1980). Great progress is being made in defining the genetic factors involved in autoimmunity and how they influence cell surface antigens and the cellular interactions that underlie immunoregulatory events. These several factors and the multiple immunopathogenetic mechanisms involved offer opportunities for new therapeutic approaches based on pathways which, although outside the classical immune system, may nevertheless restore immune regulation and ameliorate disease.

REFERENCES

Agnello, V. (1981) Detection of immune complexes. In *Current Perspectives on the Immunology of SLE*. (Sponsored by The Hannemann Medical College and Hospital, June 1981, Atlanta, GA (Ed.) Koffler, D. *Arthritis and Rheumatism* (Supplement).

Ansar Ahmed, S. & Penhale, W. J. (1980) Role of sex steroids in autoimmune thyroiditis. In *Abstracts of the 4th International Congress of Immunology*. (Ed.) Preud'Homme, J. L. & Hawken, V. A. L. 14.3.02.

Besedovsky, H. & Sorkin, E. (1977) Network of immune–neuroendocrine interactions. *Clinical and Experimental Immunology*, **27**, 1–12.

Chapel, T. A. & Burns, R. E. (1971) Oral contraceptives and exacerbations of lupus erythematosus. *American Journal of Obstetrics and Gynecology*, **110**, 366–369.

Cohn, D. A. (1979) High sensitivity to androgen as a contributing factor in sex differences in the immune response. *Arthritis and Rheumatism*, **22**, 1218–1233.

Kornreich, H. (1976) Systemic lupus erythematosus in childhood. *Clinical Rheumatology Disease*, **2**, 249–254.

Lahita, R. G., Bradlow, H. L., Kunkel, H. G. & Fishman, J. (1979) Alterations of estrogen metabolism in systemic lupus erythematosus. *Arthritis and Rheumatism*, **22**, 1195–1198.

Mund, A., Simson, J. & Rothfield, N. (1963) Effect of pregnancy on course of systemic lupus erythematosus. *Journal of the American Medical Association*, **183**, 917–920.

Murphy, E. D. & Roths (1978) Autoimmunity and lymphoproliferation: Induction by mutant gene *lpr*, and acceleration by a male-associated factor in strain BXSB. *Genetic Control of Autoimmune Disease* (Ed.) Rose, N. R. Bigazzi, P. E. & Warner, N. L. pp. 207–230. New York: Elsevier North Holland.

Rogers, M. P., Dubey, D. & Reich, P. (1979) The influence of psyche and the brain on immunity and disease susceptibility: a critical review. *Psychosomatic Medicine*, **41**, 147–164.

Rose E. & Pillsbury, D. M. (1944) Lupus erythematosus (erythematodes) and ovarian function: observations on a possible relationship, with a report of 6 cases. *Annals of Internal Medicine*, **21**, 1022–1034.

Roubinian, J. R., Papoian, R. & Talal, N. (1977) Adrenergic hormones modulate autoantibody responses and improve survival in murine lupus. *Journal of Clinical Investigation*, **59**, 1066–1070.

Roubinian, J. R., Talal, N., Greenspan, J. S., Goodman, J. R. and Siiteri, P. K. (1978) Effect of castration and sex hormone treatment on survival, anti-nucleic acid antibodies, and glomerulonephritis in NZB/NZW F_1 mice. *Journal of Experimental Medicine*, **147**, 1568–1583.

Roubinian, J. R., Talal, N., Greenspan, J. S., Goodman, J. R. and Nussenzweig, V. (1979a) Danazol's failure to suppress autoimmunity in NZB/NZW F_1 mice. *Arthritis and Rheumatism* (Brief Report), **22**, 1399–1402.

Roubinian, J. R., Talal, N. Greenspan, J. S., Goodman, J. R. & Siiteri, P. K. (1979b) Delayed androgen treatment prolongs survival in murine lupus. *Journal of Clinical Investigation*, **63**, 902–911.

Sawada, S. & Talal, N. (1979) Evidence for a helper cell promoting anti-DNA antibody production in murine lupus. (Abstract.) *Arthritis and Rheumatism*, **22**, 655–656.

Shear, H. L., Roubinian, J. R., Gil, P. & Talal, N. (1982) Clearance of sensitized erythrocytes in NZB/NZW mice: effects of castration and sex hormone treatment. *European Journal of Immunology* (in press).

Steinberg, A. D., Melez, K. A., Raveche, E. S., Reeves, J. P., Boege, W. A., Smathers, P. A., Taurog, J. D., Weinlein, L. Duvic, M. (1979) Approach to the study of the role of sex hormones in autoimmunity. *Arthritis and Rheumatism*, 22, 1170–1176.

Steinberg, A. D., Roths, J. B., Murphy, E. D., Steinberg, R. T. & Raveche, E. S. (1980) Effects of thymectomy or androgen administration upon the autoimmune disease of MRL/Mp-lpr/lpr mice. *Journal of Immunology*, **125**, 871–873.

Stern, R., Fishman, J., Brusman, H. & Kinkel, H. G. (1977) Systemic lupus erythematosus associated with Klinefelter's syndrome. *Arthritis and Rheumatism*, **20**, 18–22.

Stumpf, W. E. & Sar, M. (1976) Receptors and mechanism of action of steroid hormones. In *Modern Pharmacology–Toxicology* (Ed.) Pasqualini, J., pp. 41–84. New York; Marcel Dekker.

Talal, N. (1979) Systemic lupus erythematosus, autoimmunity, sex and inheritance. (Editorial) *New England Journal of Medicine*, **301**, 838–839.

Talal, N., Roubinian, J. R., Shear, H., Hom, J. T. & Miyasaka, N. (1980) Progress in the mechanisms of autoimmune disease. *Immunology 1980 (4th International Congress of Immunology, Paris, France, 1980)* (Ed.) Fougereau, M., London: Academic Press. **3**, 889.

Talal, N., Roubinian, J. R., Dauphinee, M. J., Jones, L. A. & Siiteri, P. K. (1981) Effects of sex hormones on spontaneous autoimmune disease in NZB/NZW hybrid mice. *Advances in Immunopharmacology (Proceedings of the 1st International Congress of Immunopharmacology, Brighton, England, 1980)*. (Ed.). Hadden, J. et al, pp. 127–137. Oxford & New York: Pergamon Press.

Theofilopoulos, A. N., Eisenberg, R. A., Bourdon, M., Crowell, Jr, J. S. & Dixon, F. J. (1979) Distribution of lymphocytes identified by surface markers in murine strains with systemic lupus erythematosus-like syndromes. *Journal of Experimental Medicine*, **149**, 516–534.

Theofilopoulos, A. N., Shawler, D. L., Eisenberg, R. A. & Dixon, F. J. (1980) Splenic immunoglobulin-secreting cells and their regulation in autoimmune mice. *Journal of Experimental Medicine*, **151**, 446–466.

Wybran, J., Appelboom, T., Famaey, J-P. and Govaerts, A. (1979) Suggestive evidence for receptors for morphine and methionine-enkephalin on normal human blood T lymphocytes. *Journal of Immunology*, **123**, 1068–1070.

Yocum, M. W., Grossman, J., Waterhouse, C., Abraham, G. N., May, A. G. & Condemi, J. J. (1975) Monozygotic twins discordant for systemic lupus erythematosus. *Arthritis and Rheumatism*, **18**, 193–199.